

geeksforgeeks.org/problems/the-painters-partition-problem1535/1

Search...

Get 90% Refund

Courses

Tutorials

Practice

Jobs

Problem

Editorial

Submissions

Comments

Output Window

Compilation Results

Custom Input

Y.O.G.I. (AI Bot)

Problem Solved Successfully

Suggest Feedback

Test Cases Passed

1112 / 1112

Attempts : Correct / Total

1 / 1

Accuracy : 100%

Points Scored

8 / 8

Your Total Score: 26

Time Taken

0.46

Solve Next

Partition array to k subsets

Allocate Minimum Pages

Longest Increasing Path in a Matrix

C++ (17)

Start Timer

```
1 class Solution {
2 public:
3     bool ispos(vector<int>& arr, int k, int maxTime) {
4         int painters = 1;
5         int time = 0;
6         for (int i = 0; i < arr.size(); i++) {
7             if (time + arr[i] <= maxTime) {
8                 time += arr[i];
9             }
10            else {
11                painters++;
12                time = arr[i];
13            }
14            return painters <= k;
15        }
16    }
17    int minTime(vector<int>& arr, int k) {
18        int sum = 0;
19        int maxVal = INT_MIN;
20        for (int i = 0; i < arr.size(); i++) {
21            sum += arr[i];
22            maxVal = max(maxVal, arr[i]);
23        }
24        int l = maxVal;
25        int r = sum;
26        int ans = 0;
27        while (l <= r) {
28            int mid = l + (r - l) / 2;
29            if (ispos(arr, k, mid)) {
30                ans = mid;
31                r = mid - 1;
32            }
33            else {
34                l = mid + 1;
35            }
36        }
37        return ans;
38    }
39};
```